Velocity graphs and components

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Read a velocity graph and describe motion in two directions.

SUCCESS CRITERIA - I can...

1. Use signed velocity-time area to find displacement.
2. Represent horizontal and vertical components separately.
3. Distinguish graph height, slope and vector magnitude.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart is at **4**, **10**, **16 m** at **0**, **1**, **2 s**.
1. **Predict its position** at **8 s**.
2. **State something you** are treating as true and a measurement that could challenge it.

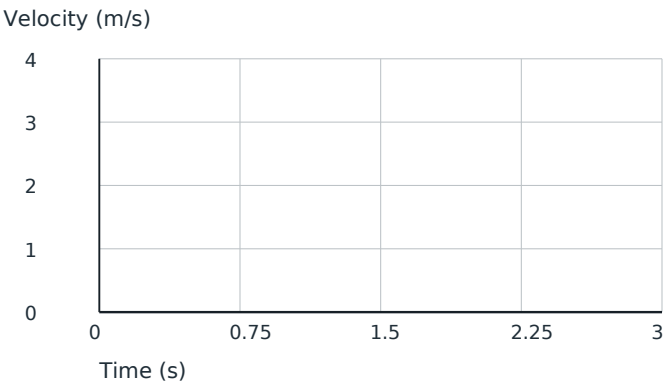
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = v_x \Delta t$; $\Delta y = v_y \Delta t$.
For constant components. Subscripts x and y name the horizontal and vertical directions. Signed velocity-time area gives displacement.

Worked reasoning: A cart travels at **3 m/s** *east* for **2 s**. The rectangle under its velocity-time graph has area **3×2=6 m**, its displacement.

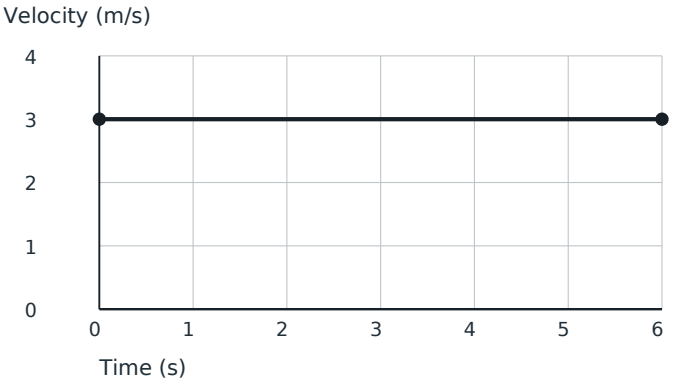
Your turn: Draw the cart's constant velocity on the axes. **Shade the area** between the line and the time axis. **Explain why multiplying the two** axis units gives meters.



Marzano check - circle: ① ② ③

03 Steady does not mean stopped

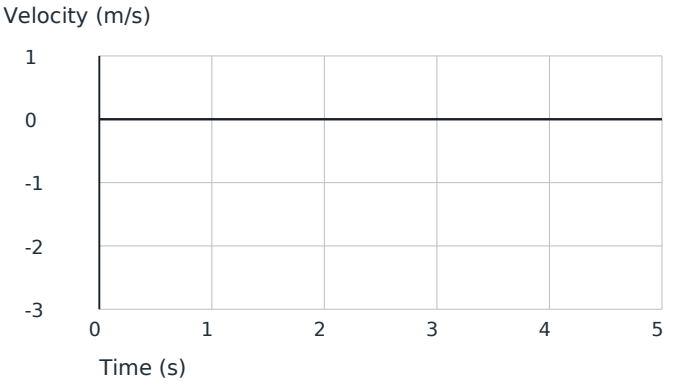
A remote-control car follows the velocity-time graph for **6 s**. Is it stopped, moving steadily, or speeding up? **Shade the area under the** line and **find its displacement**.



Marzano check - circle: ① ② ③

04 Move, then wait

A robot drives *west* at **2 m/s** for **3 s**, then waits for **2 s**.
1. **Draw its velocity-time graph**, taking *east* as positive.
2. **How far *west*** is it from the start?
3. **Explain why waiting** does not undo the trip.



Marzano check - circle: ① ② ③

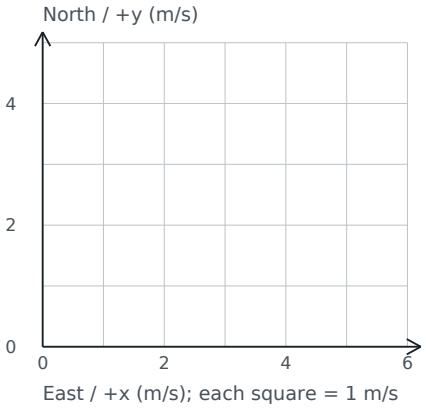
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Continue down each column.

05 One diagonal motion

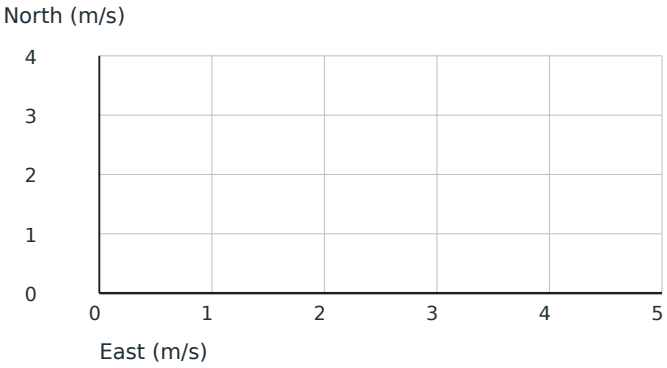
A drone moves **4 m** *east* and **3 m** *north* during each second of flight. These motions happen together. **Draw the velocity components** on the grid and **label each** in m/s.



Marzano check - circle: 1 2 3

06 Make the components visible

For the drone in question **5**, the diagonal velocity arrow has a supplied length of **5 m/s**. A student adds **4** + **3** and reports **7 m/s**. Use the vector triangle to **explain why** that addition does not give the speed.



Marzano check - circle: 1 2 3

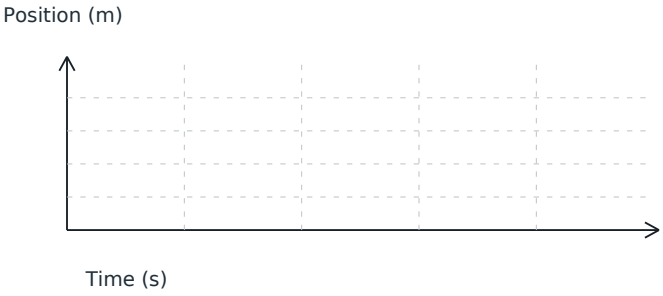
07 Fix the statement about speed

For a velocity with components **4 m/s** *east* and **3 m/s** *north*, the diagonal speed is given as **5 m/s**. A student says it is **7 m/s**. Explain using a sketch.

Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A cart goes from **15 m** to **5 m** in **3 s** at steady velocity. **Find velocity** and **sketch its position-time graph**. **Label the diagram** with the given positions and any prediction.



Marzano check - circle: 1 2 3

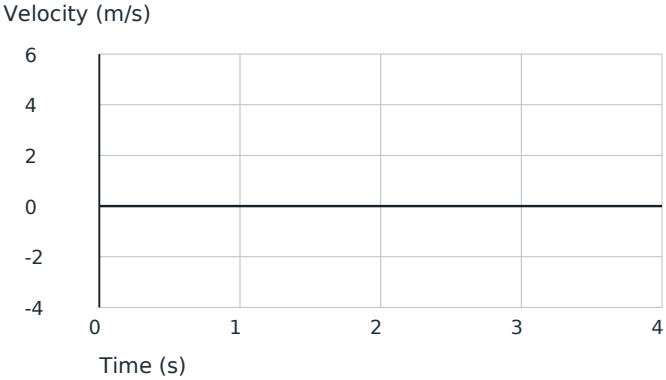
09 Exit check

A velocity graph stays at **-2 m/s** for **5 s**. **Give displacement** with direction. A different object has components **2 m/s** *east* and **1 m/s** *south*: **sketch its motion direction**.

Marzano check - circle: 1 2 3

10 STRETCH: test the prediction

1. **Draw a four-second velocity-time graph** with nonzero motion but zero displacement.
2. **Give its distance traveled**.



Marzano check - circle: 1 2 3